

Query corregido

Como el **CREATE TABLE** anterior falló, probablemente la tabla no existe, pero igual incluiría el **DROP** para asegurarnos:

```
sdf_sql(  
  sc,  
  "  
  DROP TABLE IF EXISTS bu_frd.eda_credit_age_2022_2025  
  "  
)
```

Luego:

```
sdf_sql(  
  sc,  
  "  
  CREATE TABLE bu_frd.eda_credit_age_2022_2025  
  STORED AS PARQUET  
  AS  
  
  WITH cohort AS (  
  
    SELECT  
      party_id,  
      observation_date,  
      observation_year,  
      label_3m  
  
    FROM bu_frd.model_features_ab_final_2022_2025  
  ),  
  
  credit_dates AS (  
  
    SELECT  
      f.party_id,  
      f.observation_date,  
      f.observation_year,  
      f.label_3m,  
  
      MIN(  
        CASE  
          WHEN t.open_dte IS NOT NULL  
            AND t.open_dte <= f.observation_date  
            THEN t.open_dte  
        END  
      ) AS earliest_open_dte
```

```

FROM cohort f

LEFT JOIN bu_frd.model_trades_behavioral_final_13m_2022_2025 t

    ON f.party_id = t.party_id
    AND f.observation_date = t.observation_date
    AND t.data_date = f.observation_date

GROUP BY
    f.party_id,
    f.observation_date,
    f.observation_year,
    f.label_3m
)

SELECT
    party_id,
    observation_date,
    observation_year,
    label_3m,
    earliest_open_dte,

CASE
    WHEN earliest_open_dte IS NOT NULL
    THEN CAST(
        FLOOR(
            MONTHS_BETWEEN(
                observation_date,
                earliest_open_dte
            )
        ) AS INT
    )
END AS credit_age_months

FROM credit_dates
"
)

```

Qué estamos midiendo exactamente

Conviene fijar desde ahora la terminología para el Word:

Credit Age: number of months between the observation date and the opening date of the oldest observable trade available for that party at the observation-date snapshot.

Matemáticamente:

$$CreditAge_i = ObservationDate_i - \min (OpenDate_{ij})$$

Esto es distinto de:

- **12-month behavioral lookback** → ventana usada para construir las features.
- **Customer tenure** → no disponible de forma confiable.
- **data_date** → snapshot/source availability date, no tenure.

Por qué me gusta más esta solución que volver ahora a raw **trades**

Sabemos que la tabla raw sí contiene más campos, incluido **frst_rpt_dte**, pero es enorme. Para este requirement no necesitamos volver a procesar cientos de miles de millones de registros si ya tenemos un snapshot point-in-time validado con **open_dte**.

Además, para el requerimiento:

“if victims at month 24 of history, then also non victims at that same age”

podemos responder perfectamente utilizando:

$$credit_age_months = 24$$

basado en age of oldest trade.

Antes de continuar con todos los análisis: dos QA muy importantes

Una vez creada la tabla, primero ejecutaría estos.

QA 1 — Integridad

```
credit_age_integrity_qa <- sdf_sql(
  sc,
  "
  SELECT
    COUNT(*) AS total_rows,

    COUNT(DISTINCT party_id)
    AS distinct_parties,

    SUM(
      CASE WHEN label_3m = 1
        THEN 1 ELSE 0 END
```

```

) AS positives,

SUM(
  CASE WHEN label_3m = 0
    THEN 1 ELSE 0 END
) AS negatives

FROM bu_frd.eda_credit_age_2022_2025
"
) %>%
collect()

```

credit_age_integrity_qa

Esperamos exactamente:

Metric	Expected
total_rows	688,589
distinct_parties	688,589
positives	62,599
negatives	625,990

QA 2 — Coverage

```

credit_age_coverage <- sdf_sql(
  sc,
  "
SELECT
  label_3m,

  COUNT(*) AS parties,

  COUNT(credit_age_months)
  AS parties_with_credit_age,

  SUM(
    CASE WHEN credit_age_months IS NULL
      THEN 1 ELSE 0 END
  ) AS missing_credit_age,

  COUNT(credit_age_months) * 1.0 /
  COUNT(*) AS credit_age_coverage

FROM bu_frd.eda_credit_age_2022_2025

```

```
GROUP BY label_3m
```

```
ORDER BY label_3m
```

```
"
```

```
) %>%
```

```
collect()
```

```
credit_age_coverage
```

Primero haría esos dos. Si salen bien, **los queries que te pasé anteriormente para `credit_age_quality`, `credit_age_summary_by_label`, `credit_age_smd`, `bands` y `exact-match validation` continúan funcionando prácticamente sin cambios**, porque todos utilizan `credit_age_months`.

Y algo importante: este error en realidad nos ayudó a simplificar la metodología. Ahora tenemos una sola definición clara y reproducible de Credit Age basada en `open_dte`, sin mezclar dos definiciones distintas.